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Peak instantaneous PlayerLoad metrics highlight movement strategy deficits in professional male soccer players

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ABSTRACT

To investigate the influence of task, limb dominance and previous injury on single leg hop task performance and loading response, 25 professional male soccer players completed anterior, medial and lateral hop tests with an accelerometer at mid-calf. Performance outcome was defined as hop distance with loading response defined as the magnitude of, and time to peak instantaneous planar PlayerLoad. The performance was sensitive to task and previous injury ($P < 0.001$) but not limb dominance, with no evidence of bilateral asymmetry ($P = 0.668$). Despite impaired performance, previously injured players did not exhibit lower peak instantaneous PlayerLoad after impact in any plane ($P \geq 0.110$). There was however a significantly ($P = 0.001$) longer time to peak medio-lateral loading after impact in previously injured players' affected limb. This observation was exacerbated when the injury was to the non-dominant limb ($P = 0.041$). Lower-limb accelerometry enhances understanding of movement strategy beyond task outcome, with practical implications in player screening and objective rehabilitation.

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KEYWORDS
Accelerometry; soccer; rehabilitation; profiling; 1D PlayerLoad

Introduction

Player profiling is systematically adopted in elite professional soccer to inform practice regarding player readiness and injury risk. Efficacy is influenced by the sensitivity of the test to factors such as previous injury, the greatest risk factor for subsequent injury (Hagglund et al., 2006). A recent analysis of counter-movement jump testing in elite soccer revealed that task outcome was not sensitive to previous injury, but that this masked alterations in movement strategy (Cohen et al., 2014; Hart et al., 2019). An elite athlete might therefore be able to achieve a performance outcome, indicative of player readiness, but with movement strategy compensations suggesting that functional recovery is not complete. This is particularly true in double-leg tasks such as the double legged counter movement jump, where bilateral movement compensations can mask the latent influence of previous injury (Baumgart et al., 2017; Jordan et al., 2015). In contrast, single-leg tasks negate the opportunity for bilateral compensations and offer the potential to examine cross-contamination following injury (Mitchell et al., 2021), where the unaffected limb is also functionally impaired by the time away from sport.

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The single-leg hop for distance is the most frequently reported test of lower limb functional performance (Kotsifaki et al., 2021) but tasks have evolved from the anterior hop to diversify functional challenges, with multi-directional tasks often included (Dingenen et al., 2019; Negrete et al., 2021). Acknowledging the importance of movement strategies over gross measures of task outcome (Heishman et al., 2019; Mitchell et al., 2021), the objective assessment of hopping tasks in injury screening has often employed laboratory methodologies such as three-dimensional kinematic analysis, force platforms and electromyography. However, the accessibility to such techniques and the opportunity to develop interventions based on data such as joint moments or muscle activity is often limited. Recent applications in tri-axial accelerometry have considered the influence of playing surface on a battery of functional tasks with an accelerometer placed on the lower limb to reflect injury epidemiology (Greig et al., 2019).

The aim of the current study was to consider the efficacy of lower-limb accelerometry during a battery of single leg hopping tasks that form part of a habitual screening process within a professional soccer team. Loading response was quantified using instantaneous PlayerLoad (a derivative of changes in acceleration). Vertical force is often measured as a marker of injury risk (Padua & DiStefano, 2009), but the multi-planar mechanism of injury warrants a multi-planar analysis of impact acceleration. Therefore, whilst accumulated PlayerLoad has been associated with injury risk (Barrett et al., 2016), the present study quantifies the temporal pattern of instantaneous PlayerLoad given observations of peak ACL strain (Shin et al., 2007; Withrow et al., 2006) and peak ankle inversion up to 80msec after touchdown (Chu et al., 2010; Mok et al., 2011). The current study examined the influence of task specificity, previous injury and limb dominance within a cohort of elite, male soccer players.

Methods

Participants

Twenty-five professional male soccer players (age 23.6 ± 5.2 yrs, weight 79.2 ± 2.6 kg, professional playing history 6.7 ± 4.1 years) contracted to an English Championship league club completed the study. According to the Helsinki Declaration, all players provided written consent. Players from the same club (n = 11) were excluded from the study only on the basis that they were deemed unable to complete the testing battery. Participants identified their dominant limb, defined as their preferred kicking leg and reflecting limb dominance as a risk factor for injury (Brown et al., 2009; Dos'Santos et al., 2019; Mokhtarzadeh et al., 2017). Club medical staff retrospectively provided injury history details for each participant with players subsequently stratified into non-injured (n = 12) or previously injured (n = 13). For the previously injured players the affected limb was recorded to inform the subsequent statistical model.

Data collection

This study was completed as part of a habitual screening process and thus all players were familiar with all experimental tasks. Data were collected during a single testing session, following a standardized warm-up comprising running of varied intensity, dynamic

stretching and familiarization trials of the experimental tasks. Each participant then completed three single legged hopping tasks: a forward hop, lateral hop, and medial hop. Each participant completed the tasks in a randomized order, with three trials on each limb, in alternate order. There was 15 seconds rest between each trial, and 60 seconds rest before commencing each new hopping task to allow for verbal description and demonstration of the new task. Each hop was completed with the aim of producing maximum distance, and the landing was fixed with the hop distance recorded. If the participant failed to control the landing, lost balance or the contralateral foot made contact with the ground, the trial was not recorded. For the lateral and medial hops a straight line was marked out perpendicular to the forward hop, and if the landing was not on this line the trial was not recorded.

During the experimental trials, each participant was fitted with a triaxial accelerometer (Optimeye X4, catapult Sports, Melbourne, Australia) situated at the superior musculotendinous junction of the achilles tendon on both limbs and secured with zinc-oxide tape applied horizontally around the limb. Attachment of the accelerometer was completed by the same club physiotherapist for each player. Orientation of the accelerometer was consistent with that typically adopted at the cervico-thoracic junction so that axial planes were defined consistently with previous literature (Greig et al., 2019). Planar acceleration was collected at a sampling frequency of 100 Hz.

Data analysis

For the anterio, medial and lateral hop tasks, the trial eliciting the greatest hop distance was selected for analysis. Independent variables were defined as hop task (anterior, lateral, medial), limb dominance, and previous injury (non-injured, injured non-affected limb, injured affected limb). Dependent variables were defined as: hop distance (performance outcome measure); the peak instantaneous PlayerLoad after landing in the anterio-posterior, medio-lateral and vertical planes; and the time to peak PlayerLoad after landing in each plane. The instance of landing was defined as the peak instantaneous vertical acceleration and the uni-axial PlayerLoad was calculated based on the rate of change of acceleration, as described by Greig et al. (2019).

Univariate general linear modelling was used to investigate a main effect for hop task, limb dominance, and previous injury in each dependent variable. Task, dominance and previous injury interactions were also examined. Where appropriate post-hoc analyses were applied to determine the directional main effect, with statistical significance predetermined at $P \leq 0.05$. All data are subsequently presented as mean ± standard deviation.

Results

Performance

Figure 1 summarizes the influence of hopping task, limb dominance and previous injury on the performance outcome measure of hop distance.

There was a significant main effect for task ($P < 0.001$) with performance in the anterior hop (192.4 ± 16.9 cm) significantly greater than the medial (160.1 ± 18.4 cm) and lateral (156.4 ± 21.1 cm) hops, which were themselves not different ($P = 0.433$).

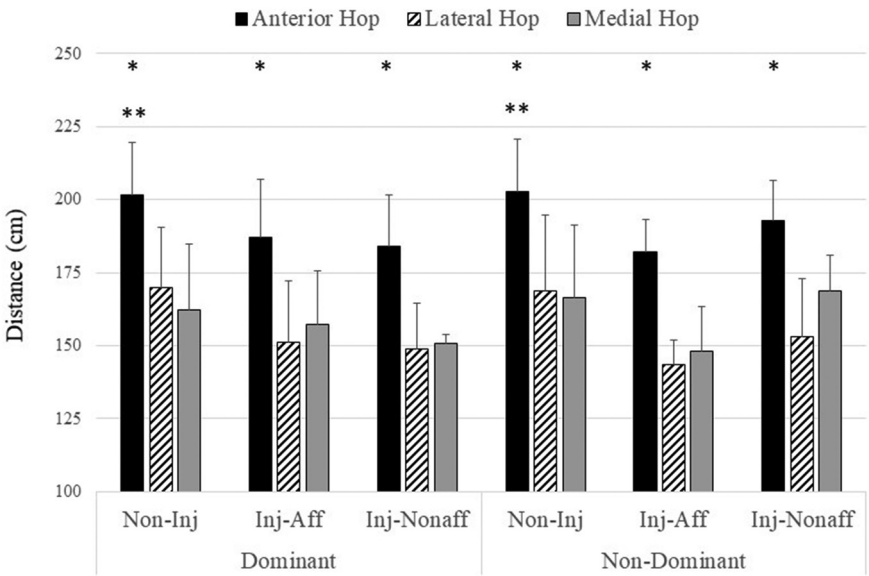

Figure 1. The influence of task, limb dominance and previous injury on hop performance. * denotes significant main effect for task; ** denotes significant main effect for previous injury.

There was also a significant main effect for previous injury (P < 0.001), with the non-injured group (178.4 ± 26.7 cm) hopping further than the injured non-affected (167.5 ± 21.6 cm) and injured affected limb (162.3 ± 23.3 cm), which were themselves not different (P = 0.257). There was no significant task *x* injury interaction (P = 0.730).

There was no significant main effect for limb dominance (P = 0.668), and no significant dominance *x* injury (P = 0.123), dominance *x* task (P = 0.744), or dominance *x* injury *x* task (P = 0.953) interactions.

Peak instantaneous playerLoad

Figure 2 summarizes the influence of task, previous injury and limb dominance on peak instantaneous vertical PlayerLoad.

There was a significant main effect for task (P = 0.003) with the anterior hop (0.14 ± 0.06 a.u) eliciting significantly higher vertical PlayerLoad than the lateral (0.06 ± 0.04 a.u) or medial (0.08 ± 0.04 a.u) hops, which were themselves not different (P = 0.628).

Peak vertical PlayerLoad was significantly (P = 0.045) lower in the injured affected-limb (0.06 ± 0.04 a.u) than the injured non-affected limb (0.11 ± 0.05 a.u), which were not different (P ≥ 0.119) to the non-injured group (0.09 ± 0.04 a.u). There was no significant task *x* injury interaction (P = 0.122).

There was no significant main effect for limb dominance (P = 0.613), and no significant task *x* dominance (P = 0.974), injury *x* dominance (P = 0.122), or task *x* injury *x* dominance (P = 0.096) interaction.

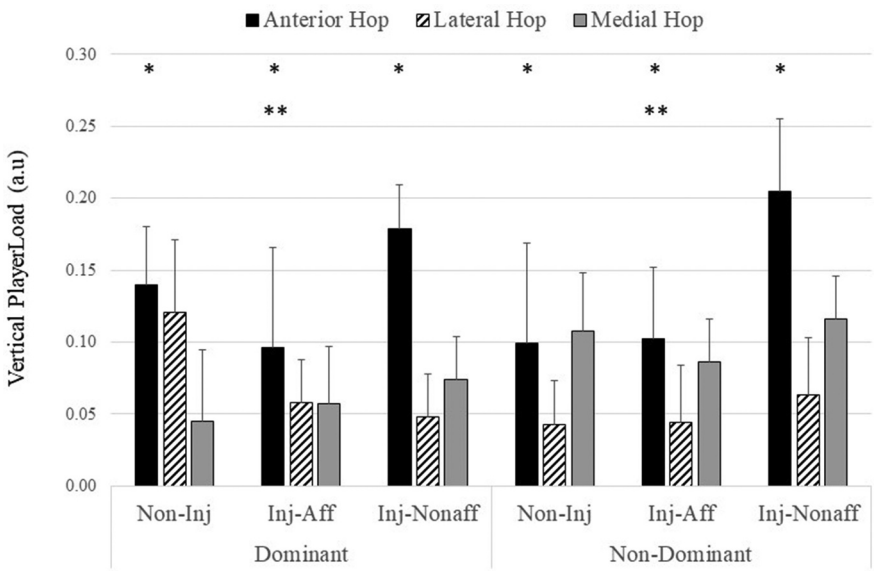

Figure 2. The influence of task, limb dominance and previous injury on the magnitude of peak instantaneous vertical PlayerLoad. * denotes significant main effect for task; ** denotes significant main effect for previous injury.

Figure 3 highlights that there was also a significant main effect for task (P = 0.004) in antero-posterior PlayerLoad at touchdown, with the anterior hop (0.12 ± 0.06 a.u) eliciting significantly higher vertical PlayerLoad than the lateral (0.06 ± 0.05 a.u) or medial (0.07 ± 0.05 a.u) hops, which were themselves not different (P = 0.634).

There was no significant main effect for injury (P = 0.738) and no injury *x* task interaction (P = 0.540).

There was no significant main effect for limb dominance (P = 0.932), and no significant task *x* dominance (P = 0.832), injury *x* dominance (P = 0.310), or task *x* injury *x* dominance (P = 0.656) interaction.

In peak medio-lateral PlayerLoad there was no significant main effect for task (P = 0.371) or injury (P = 0.704), and no task *x* injury interaction (P = 0.741), as summarized in Figure 4.

There was a significant main effect for limb dominance, with the dominant limb (0.08 ± 0.05 a.u) eliciting significantly (P = 0.008) higher peak medio-lateral PlayerLoad than the non-dominant limb (0.05 ± 0.04 a.u). There was also a significant limb *x* task interaction (P = 0.050), with the dominant limb higher in the anterior hop but lower in the medial hop.

A significant limb *x* task *x* injury interaction (P = 0.042) revealed that the bilateral asymmetry in the medial hop was exacerbated in the previously injured players.

Time to peak instantaneous playerLoad

There was no significant main effect for task in the vertical (0.027s; P = 0.367), antero-posterior (0.019s; P = 0.918) or medio-lateral (0.025s; P = 0.461) planes. Similarly, there was no significant main effect for dominance in the vertical (P = 0.306), antero-posterior (P = 0.979) or medio-lateral (P = 0.892) planes.

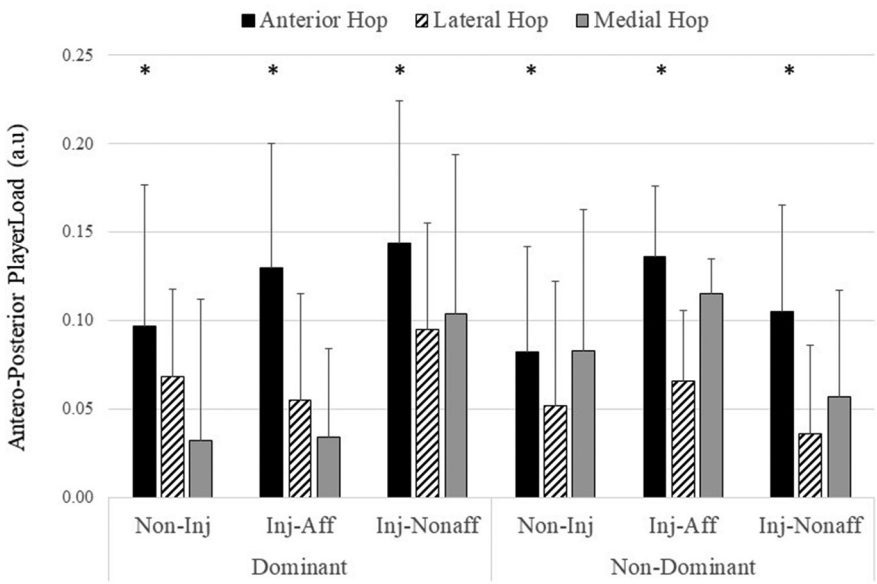

Figure 3. The influence of task, limb dominance and previous injury on the magnitude of peak instantaneous antero-posterior PlayerLoad. * denotes significant main effect for task.

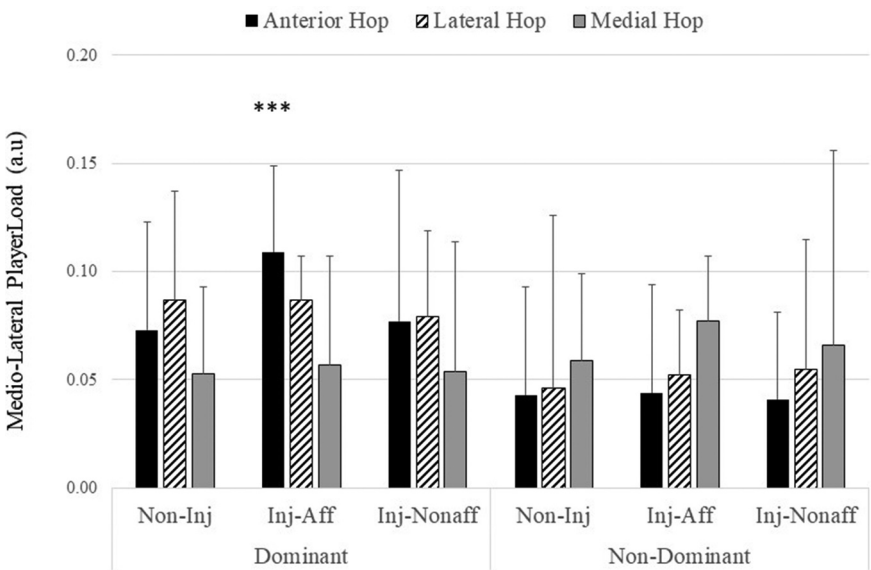

Figure 4. The influence of task, limb dominance and previous injury on the magnitude of peak instantaneous medio-lateral PlayerLoad. *** denotes significant main effect for limb.

There was no significant main effect for injury in the vertical (P = 0.258) or antero-posterior planes (P = 0.619), and no two-way or three-way interactions in these planes (P ≥ 0.365). However, in the medio-lateral plane there was a significant main effect for injury (P = 0.001), summarized in Figure 5.

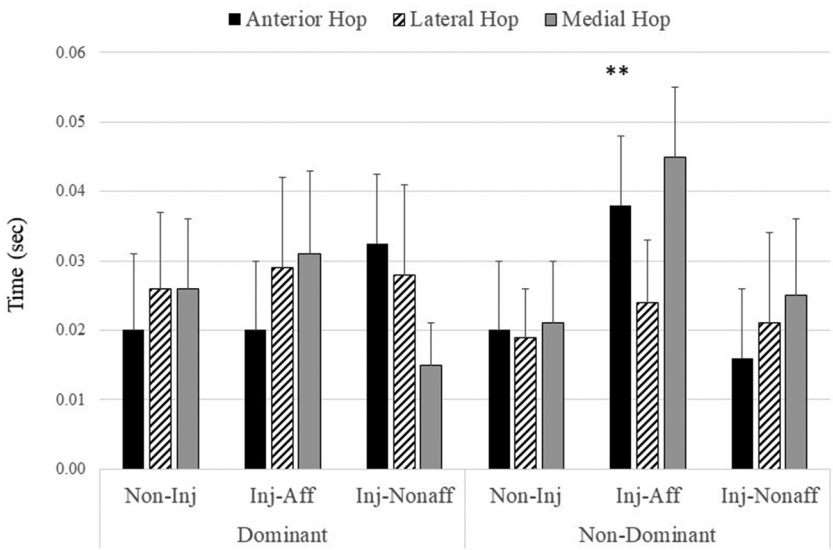

Figure 5. The influence of task, limb dominance and previous injury on the time to peak instantaneous medio-lateral PlayerLoad. ** denotes significant main effect for previous injury.

Time to peak instantaneous medio-lateral PlayerLoad was significantly longer in the injured affected limb (0.031 ± 0.013 sec) than the injured non-affected (0.022 ± 0.012 sec) or non-injured (0.022 ± 0.010 sec) limbs, which were themselves not different. A significant injury *x* limb interaction (P = 0.017) showed that the main effect for injury was more pronounced in the non-dominant limb, and further exacerbated in the medial and anterior hops with a significant injury *x* limb *x* task interaction (P = 0.041).

Discussion

The aim of the present study was to examine the sensitivity of lower limb accelerometry to task-specificity, limb dominance and previous lower limb musculoskeletal injury. As expected, the anterior hop produced a greater hop distance than the medial or lateral hop. By association, the peak instantaneous PlayerLoad in the vertical and antero-posterior planes was also greatest in the anterior hop landing, with similar loading magnitudes elicited by the medial and lateral hop landings. The greater vertical loading response reflects the greater hop distance, and potentially vertical displacement, with the greater antero-posterior loading indicative of the directional forward hopping task. In contrast, medio-lateral PlayerLoad showed no main effect for task. This is indicative of the reduced demand on medio-lateral stabilization in the anterior hop landing, despite the increased hop distance.

The task-specific response in peak instantaneous planar PlayerLoad therefore reflected the directional challenge and outcome measure of hop distance. This suggests efficacy in the use of peak instantaneous PlayerLoad in the monitoring of landing mechanics, for example, during rehabilitation. The more commonly used cumulative total PlayerLoad metric is typically developed over the duration of match-play, and as a vector sum of each plane. This metric negates examination of the mechanism of injury and provides no detail on relative planar contributions to loading. Whilst previous research has tried to associate

PlayerLoad with injury risk in sport (Barrett et al., 2016; Cummins et al., 2019), the instantaneous PlayerLoad reflecting instantaneous changes in planar acceleration at a relatively high sampling frequency of 100 Hz might offer greater clinical value.

The time to peak instantaneous PlayerLoad was also quantified, given observations of the temporal pattern of injury risk (Chu et al., 2010; Mok et al., 2011). In the current study the time to peak instantaneous PlayerLoad was not sensitive to task in any plane, but the average magnitude across all tasks and planes at ~30msec supports observations of peak ACL strain (Shin et al., 2007; Withrow et al., 2006) and ankle inversion (Chu et al., 2010; Mok et al., 2011) after touchdown, reflecting the accelerometry location at mid-calf.

The present study also considered the sensitivity of these outcome measures to limb dominance, consistently identified as an aetiological risk factor for injury (Brown et al., 2009; Dos'Santos et al., 2019; Mokhtarzadeh et al., 2017). The gross outcome measure of hop distance was not sensitive to limb dominance, which might reflect the relative lack of task complexity, the elite nature of the players, and the habituation to these exercises. Correspondingly, there was also no influence of limb dominance on peak instantaneous PlayerLoad in the vertical or antero-posterior planes. However, in the medio-lateral plane the dominant limb elicited significantly higher peak instantaneous PlayerLoad, with data pooled across all tasks. This would suggest a greater risk of injury to the dominant limb, reflective of epidemiological observations (DeLang et al., 2021), but might also be indicative of greater load tolerance in the dominant limb, given the relative simplicity of the tasks.

The observation of greater medio-lateral loading on the dominant limb was also subject to a significant limb x task interaction. The main effect for limb dominance was reflected in the anterior hop. The greater medio-lateral loading elicited by the anterior hop might be indicative of an altered landing strategy to arrest forward momentum, but which might increase injury risk. Strategies may include greater ankle inversion or knee varus displacement, as observed by Greig (2019) in the hop landing strategies of elite male soccer players. In contrast, the dominant limb elicited lower medio-lateral load in response to the medial hop. This task is less habitual to the players, and the lower loading is indicative of a safer landing given the mechanistic relevance of the medio-lateral jerk in injurious movement patterns including ankle inversion and knee valgus displacement.

The limb asymmetry in medio-lateral loading in the medial hop landing was exacerbated by previous injury, suggesting that loading tolerance and the players' perception of the functional challenge or risk must also be considered. The non-injured players performed better than the previously injured players in terms of hop distance. This was true for the affected and unaffected limbs of the previously injured players, suggesting some cross contamination. This supports the observations of Mitchell et al. (2021) in single leg counter movement jump height, where previously injured players demonstrated a different movement strategy in the affected and unaffected limb. Previous injury is commonly acknowledged as the greatest risk factor for injury (Hagglund et al., 2006) and typically associated with greater severity (Ekstrand et al., 2020). The current study and the vertical hop employed by Mitchell et al. (2021) highlight the value of single leg hops which negate the movement compensations observed in double leg jumping tasks (Baumgart et al., 2017; Hart et al., 2019).

However, despite the impaired performance, peak instantaneous PlayerLoad in the antero-posterior and medio-lateral planes was not sensitive to previous injury. A previously injured player therefore exhibits a less economical landing outcome, with

equivalence in impact loading despite impaired performance. The instantaneous peak PlayerLoad metric further identified that the previously injured affected limb elicited lower vertical loading. This might be reflective of the lower hop distance but might also reflect an impaired capacity for loading tolerance. The previously injured limb was also characterized by a greater time to peak in medio-lateral loading, indicative of an altered landing strategy, lack of capacity to absorb load quickly, and indicative of greater injury risk. Given the functional relevance of these metrics to impact forces and the temporal pattern of risk post-impact, the association with the impaired performance highlights functional deficits that might then inform rehabilitative foci.

The current study defines instantaneous PlayerLoad, a derivative of changes in acceleration, but the magnitude of planar acceleration is likely to be as important with peak vertical ground reaction force commonly used as a marker of injury (Napier et al., 2018; Van der Worp et al., 2016). Secondary analysis therefore considered planar acceleration at landing as additional dependent variables. The magnitude of axial accelerations was consistent with ground reaction force trends, with hierarchical ordering of vertical, antero-posterior, followed by medio-lateral acceleration. Vertical acceleration at touchdown was greater in the anterior hop than in the lateral hop, reflecting observations in performance and instantaneous PlayerLoad. Impact acceleration in the antero-posterior and medio-lateral planes also reflected the previous discussion in PlayerLoad, with similar main effects for limb dominance and previous injury observed. Observations from the present study advocate a multi-planar analysis of PlayerLoad metrics in player monitoring to supplement performance outcome measures with an appreciation of movement strategy.

Care should be taken when generalizing these findings beyond the elite male soccer players used and the battery of functional tests. The number of previously injured players negated the opportunity to stratify further to specific injuries, and future research might consider the influence of specific injury types and through the rehabilitative process.

Conclusions

Tri-axial accelerometry was effective in differentiating movement task and previous injury. The hierarchical ordering of planar acceleration magnitudes at landing and task-specific influence on the magnitude of and time to peak instantaneous PlayerLoad highlight efficacy in screening. Movement strategy should be given as much attention as performance score in screening, and tri-axial accelerometry provides an efficacious, *in-vivo* methodological approach that is well suited to the clinical and elite sporting context. It is further advocated that rehabilitative screening should consider both limbs to account for cross-contamination of the unaffected limb, and single-leg tasks that provide greater functional specificity and negate bilateral movement compensations which might mask incomplete functional recovery.

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